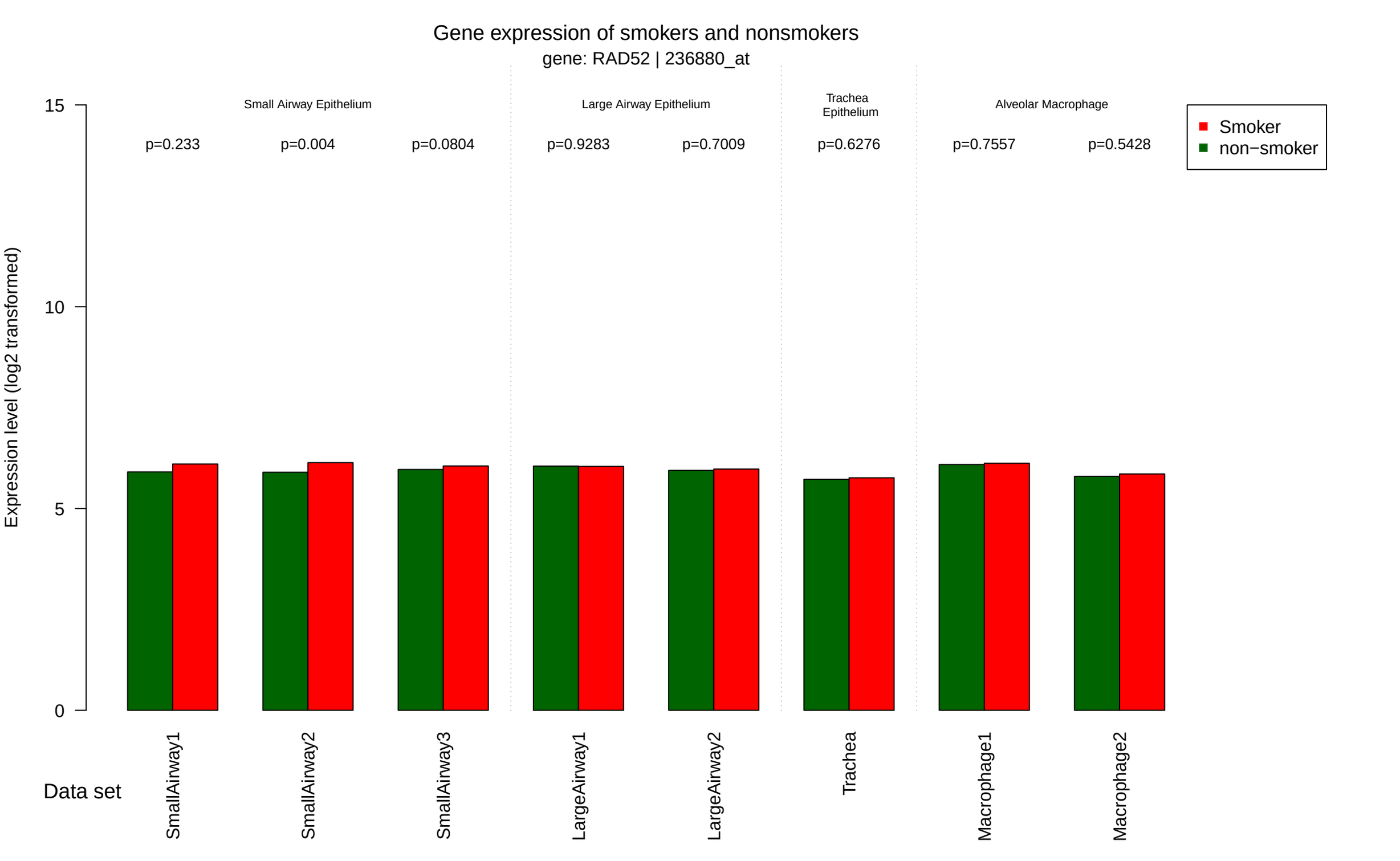
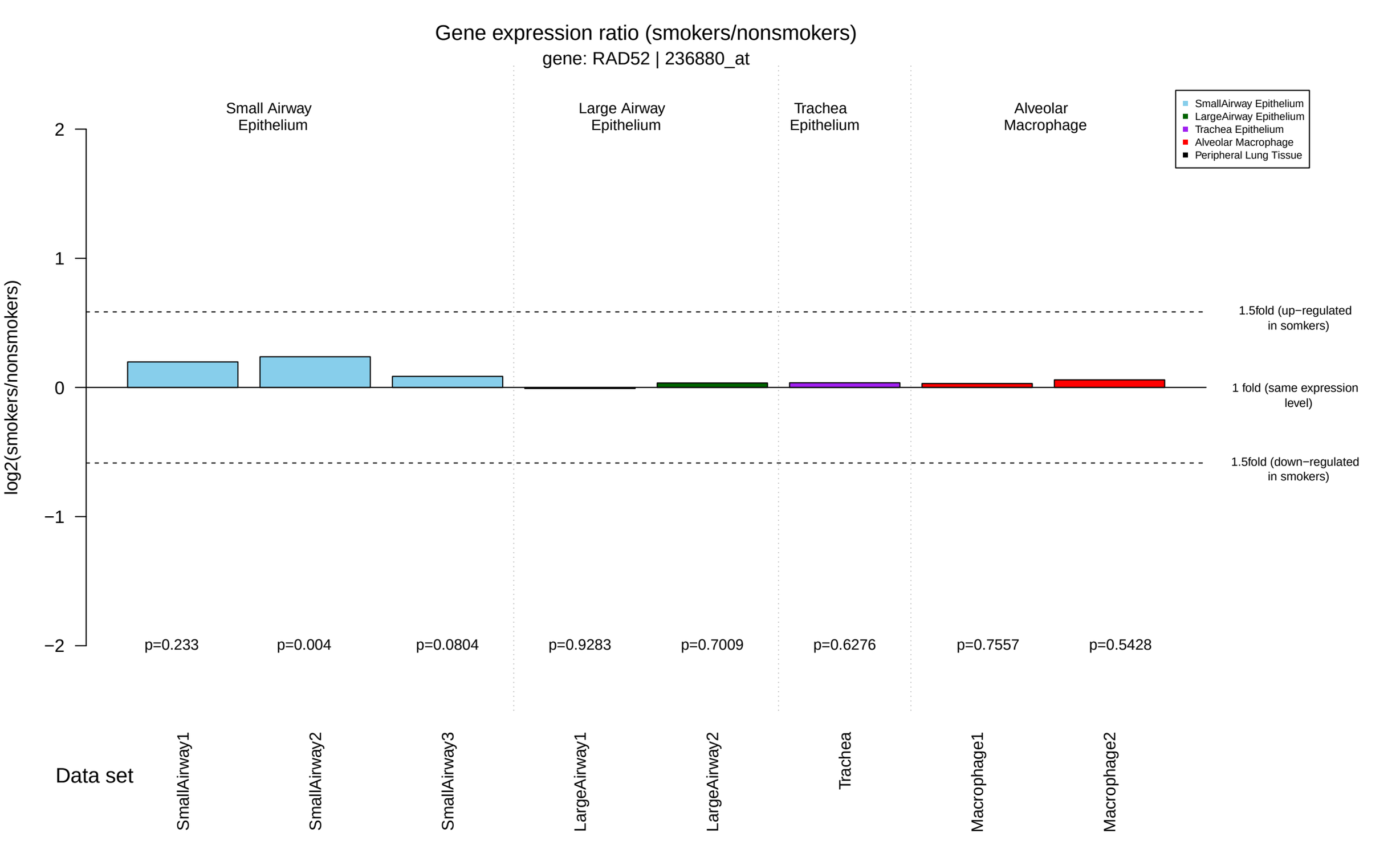
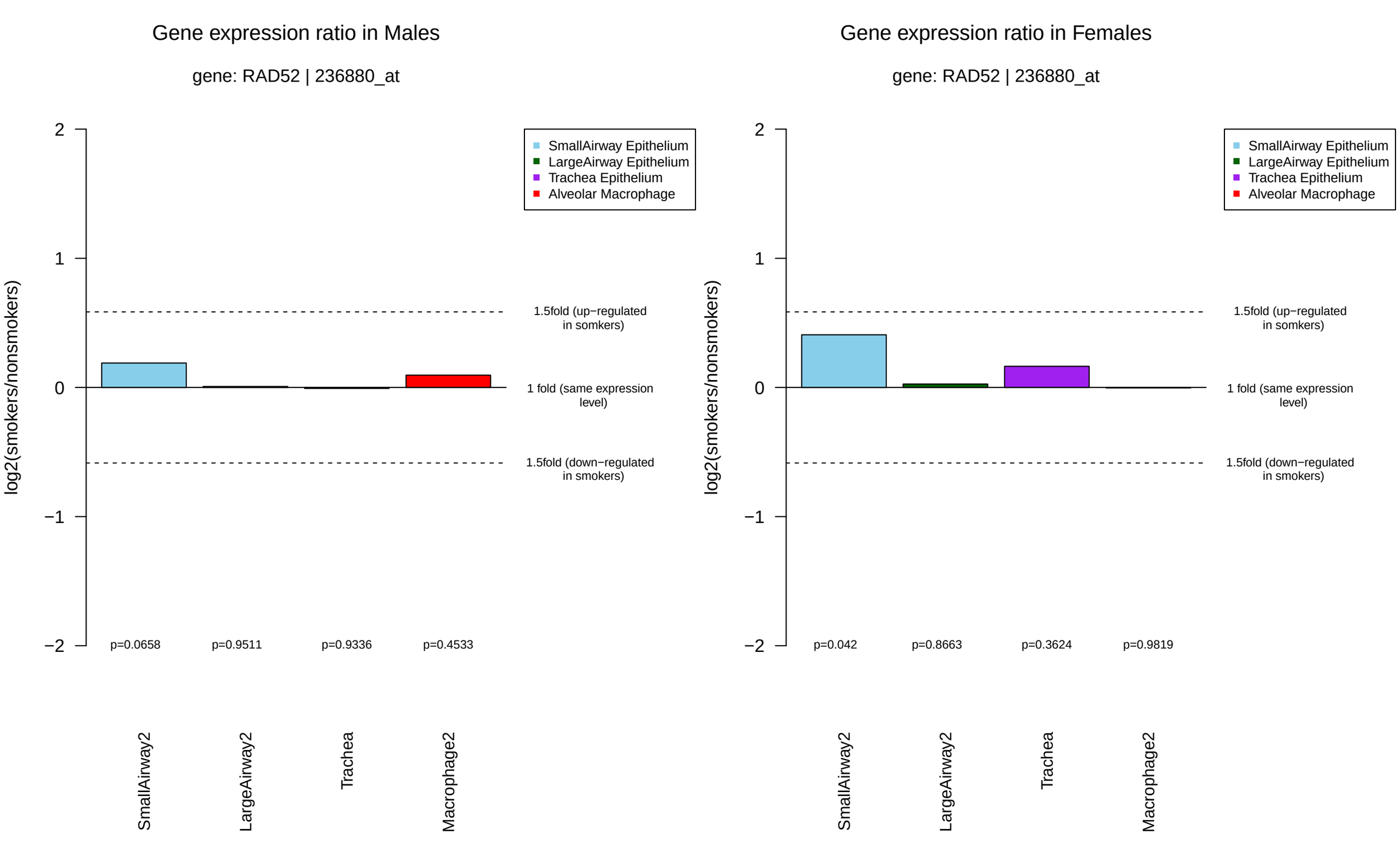
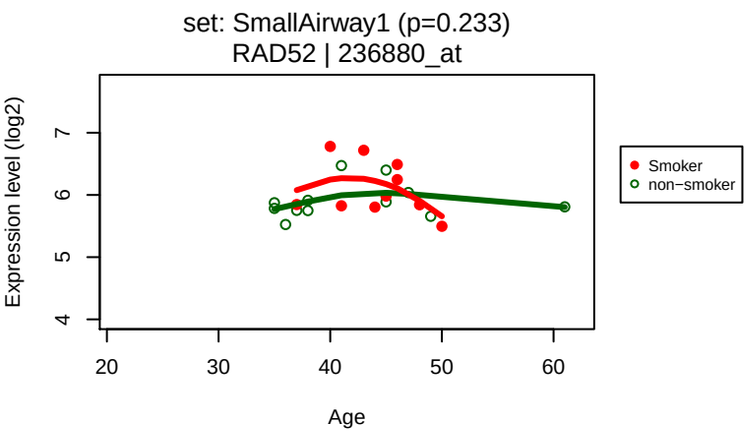
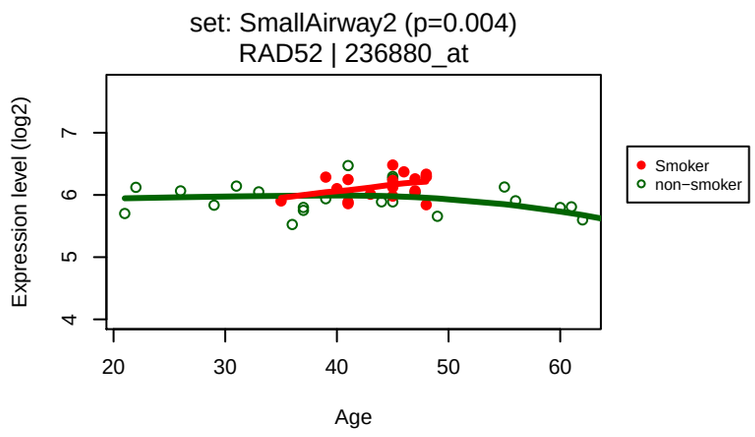
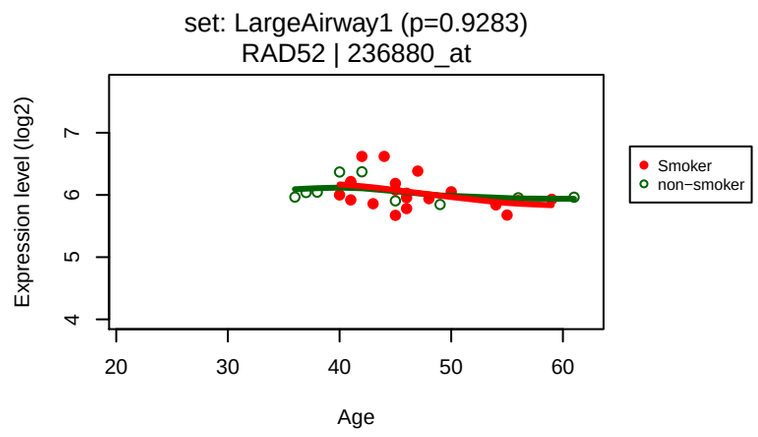

Gene expression of smokers and nonsmokers
gene: RAD52 | 236880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.233
p=0.004
p=0.0804
p=0.9283
p=0.7009
p=0.6276
p=0.7557
p=0.5428
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

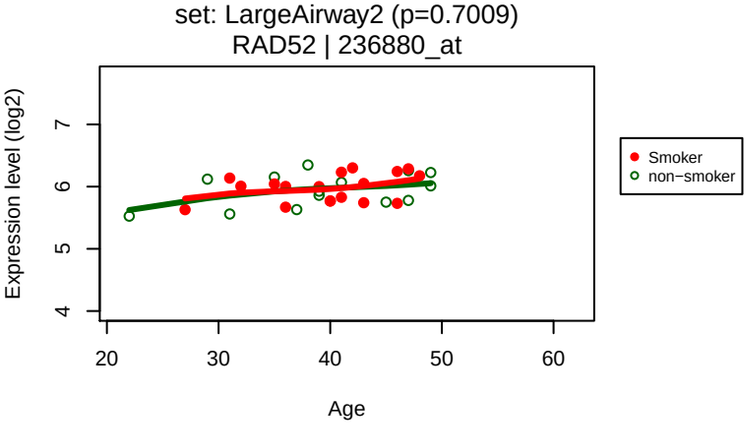

Gene expression ratio (smokers/nonsmokers)
gene: RAD52 | 236880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.233
p=0.004
p=0.0804
p=0.9283
p=0.7009
p=0.6276
p=0.7557
p=0.5428
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

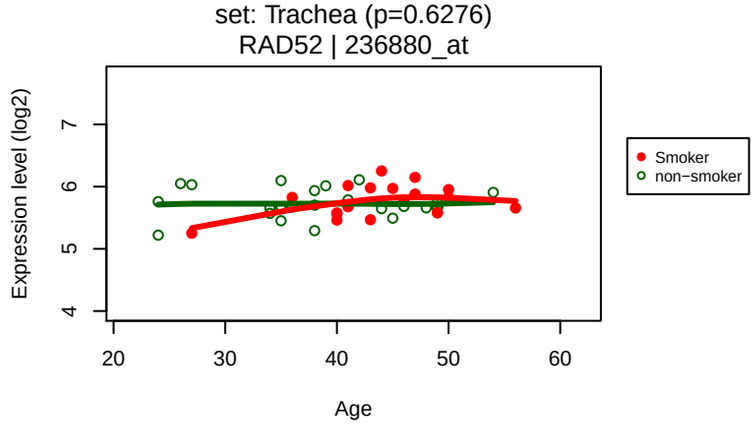

Gene expression ratio in Males
gene: RAD52 | 236880_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0658
p=0.9511
p=0.9336
p=0.4533
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD52 | 236880_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.042
p=0.8663
p=0.3624
p=0.9819
SmallAirway2
LargeAirway2
Trachea
Macrophage2

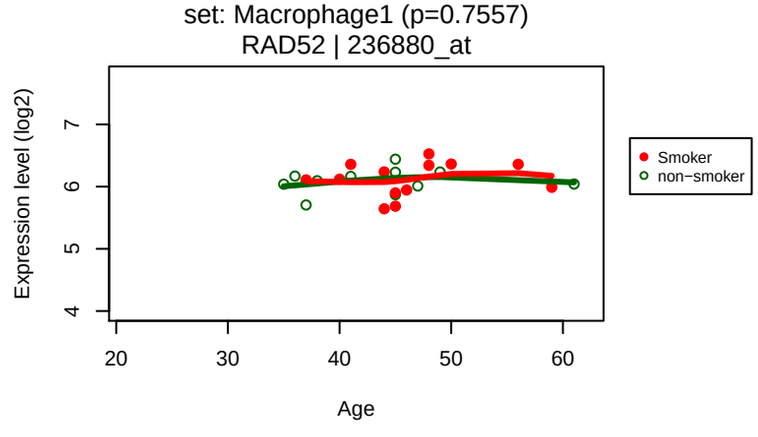

set: SmallAirway1 (p=0.233)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker

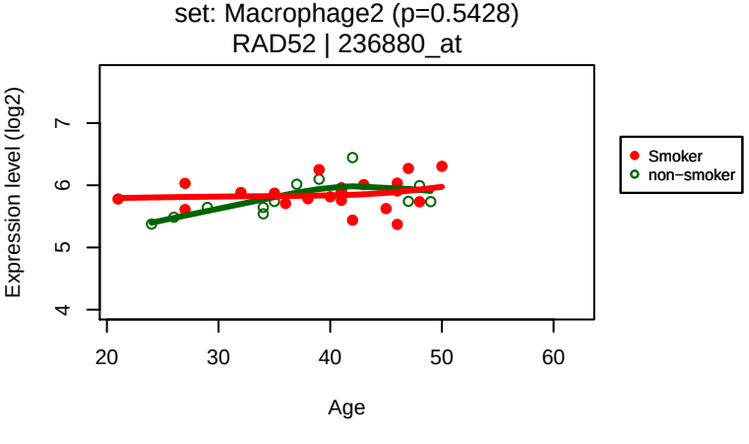

set: SmallAirway2 (p=0.004)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker

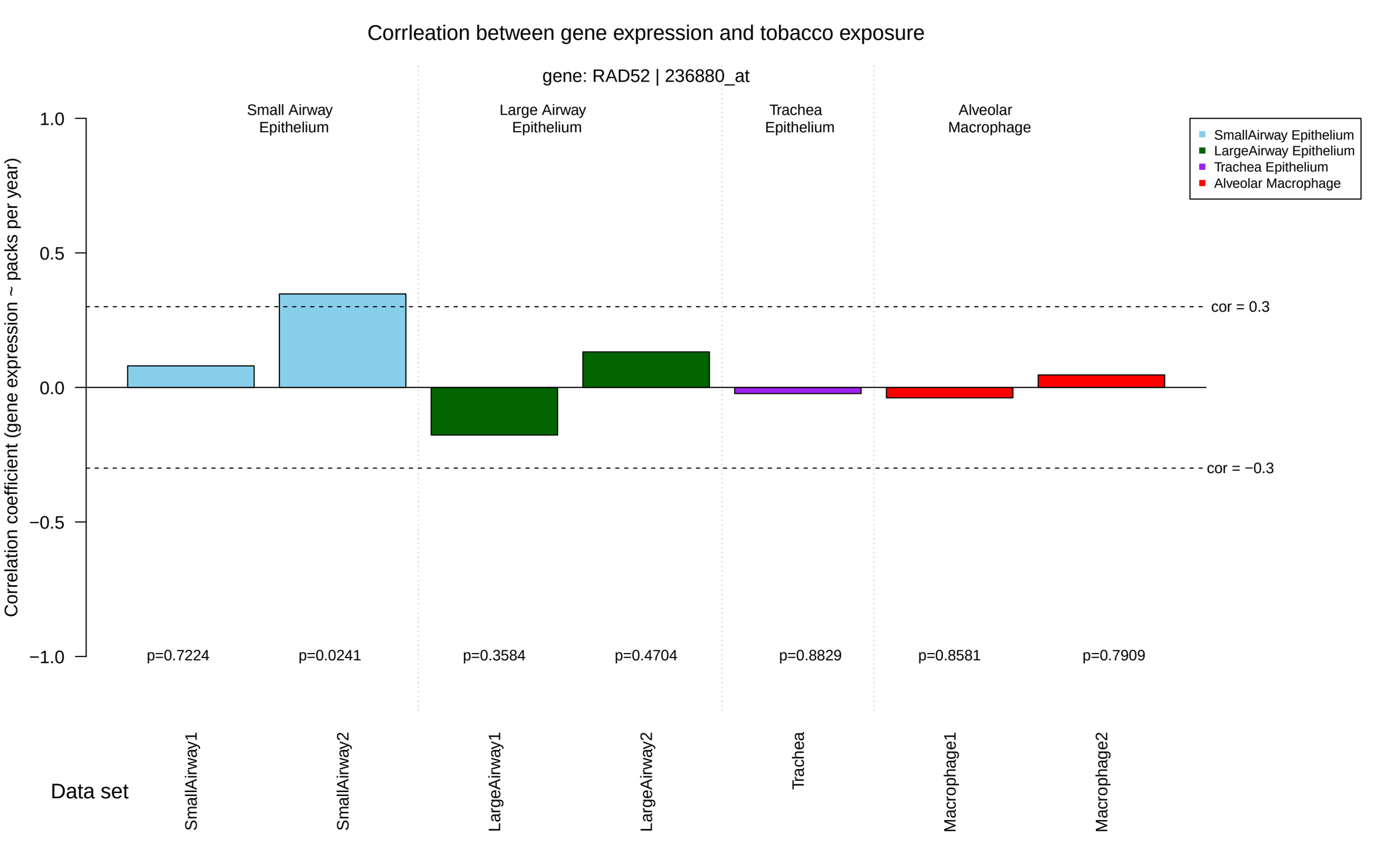

set: LargeAirway1 (p=0.9283)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7009)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6276)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7557)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5428)
RAD52 | 236880_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD52 | 236880_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7224
p=0.0241
p=0.3584
p=0.4704
p=0.8829
p=0.8581
p=0.7909
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2